Financial Fraud Detection System

Full Project Roadmap — Data Engineer Track

AI · Machine Learning · Deep Learning · MLOps

6 Weeks	5 Tracks	15 Phases	<100ms
Duration	Parallel	Milestones	Real-time

Prepared for: Data Engineering Team
Duration: 6 Weeks (42 Days) — Full Stack AI Project

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01 — Project Overview & Objectives

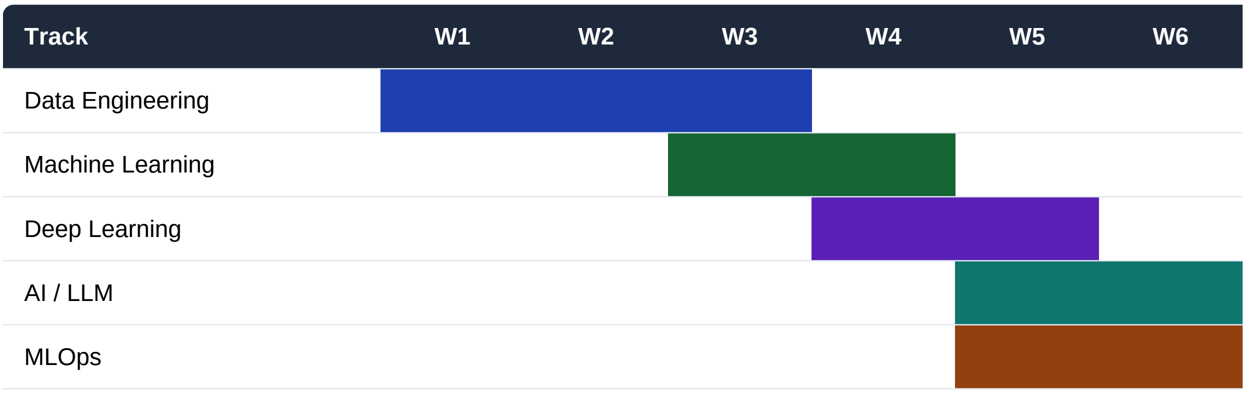
This roadmap defines a complete 6-week execution plan for building a production-grade Financial Fraud Detection System. The project spans five parallel tracks: Data Engineering, Classical Machine Learning, Deep Learning, AI/LLM integration, and MLOps. Each track has clearly defined phases, tools, and weekly deliverables.

Real-time Detection	Process and score transactions in under 100ms using streaming pipelines.
Multi-model Ensemble	Combine XGBoost, LSTM, GNN, and Autoencoder for high accuracy.
Graph Fraud Rings	Detect coordinated fraud groups using Graph Neural Networks.
LLM Explanation	Auto-generate Suspicious Activity Reports (SAR) using Claude API.
MLOps Pipeline	Full CI/CD for model training, versioning, and automated retraining.
Data Governance	Lineage, quality checks, encryption, and GDPR compliance.

02 — 6-Week Timeline at a Glance

Week 1	Data Ingestion & Architecture Setup	Kafka, Airflow, Delta Lake, S3
Week 2	ETL Pipelines & Feature Engineering	Spark, dbt, Great Expectations
Week 3	Feature Store + Classical ML Models	Feast, Redis, XGBoost, LightGBM
Week 4	Anomaly Detection + Deep Learning (LSTM)	Isolation Forest, PyTorch, LSTM
Week 5	GNN + XAI + LLM Integration	PyG, GraphSAGE, SHAP, Claude API, LangChain
Week 6	MLOps, Real-time Scoring & Production	Flink, Triton, MLflow, Grafana, CI/CD

Track Overlap — Gantt View



DE

Track 1: Data Engineering

Week 1–3

DE-1 Data Ingestion & Sources

Week 1–2

- Define data sources: transactions DB, payment APIs, device logs, KYC records
- Build Kafka topics: raw_transactions, user_events, device_signals
- Airflow DAGs for batch ingestion (daily reconciliation, historical load)
- Schema Registry setup — enforce Avro schemas & data contracts
- CDC (Change Data Capture) from core banking system via Debezium

Apache Kafka Apache Airflow Debezium Avro REST APIs

DE-2 Data Lake Architecture

Week 1–2

- S3/GCS bucket structure: bronze/ silver/ gold/ sandbox/
- Delta Lake with ACID transactions, time-travel, and Z-ordering
- Data partitioning strategy: by date, region, transaction_type
- Snowflake DWH: star schema for analytics (fact_transactions, dim_user)
- Retention policies, encryption at rest (AES-256), access control (IAM)

Delta Lake Amazon S3 Snowflake Apache Iceberg AWS IAM

DE-3 ETL Pipelines & Feature Engineering

Week 2–3

- PySpark jobs: clean, deduplicate, normalize transaction records
- dbt models: Silver layer transformations, business logic, SCD Type 2
- Feature engineering: tx_velocity_1h, avg_amount_30d, merchant_risk_score
- Time-window aggregations: rolling sums/counts over 1h, 24h, 7d, 30d
- Great Expectations data quality suite: schema, null checks, range validation
- Anomaly flagging in ETL: amount outliers, unusual hours, geo-distance spikes

Apache Spark (PySpark) dbt Core Great Expectations Pandas SQL

ML

Track 2: Machine Learning

Week 3–4

ML-1 Feature Store & Data Preparation

Week 3

- Feast Feature Store: online (Redis <5ms lookup) + offline (Parquet on S3)
- Feature groups: user_features, merchant_features, device_features, tx_features
- Class imbalance handling: SMOTE oversampling + random undersampling
- Train/validation/test split: 70/15/15 — stratified by fraud label
- Feature selection: Mutual Information, Permutation Importance, Boruta
- Baseline model: Logistic Regression to establish benchmark metrics

Feast Redis scikit-learn imbalanced-learn SMOTE

ML-2 Classical ML Models — Fraud Classification

Week 3–4

- XGBoost: gradient boosted trees, tuned scale_pos_weight for imbalance
- LightGBM: faster training on large datasets, leaf-wise growth strategy
- Random Forest: ensemble of 500 trees, feature importance analysis
- Hyperparameter tuning: Optuna with 100 trials, cross-validation (StratifiedKFold)
- Evaluation metrics: Precision@K, Recall, F1, AUC-ROC, AUC-PR, Brier Score
- Threshold optimization: maximize F1 or Recall at fixed Precision (business rule)
- Model comparison dashboard in MLflow Tracking

XGBoost LightGBM Random Forest Optuna MLflow scikit-learn

ML-3 Anomaly Detection — Unsupervised

Week 4

- Isolation Forest: detect outlier transactions without labels
- One-Class SVM: boundary learning on normal transaction distribution
- DBSCAN clustering: identify unusual spending behavior clusters
- Autoencoder reconstruction error: flag high-error transactions as anomalies
- Ensemble scoring: combine supervised + unsupervised scores (weighted average)
- Threshold calibration: ROC analysis on validation set

Isolation Forest One-Class SVM DBSCAN PyTorch Autoencoder scikit-learn

DL

Track 3: Deep Learning

Week 4–5

DL-1 Sequence Models — Temporal Patterns

Week 4

- LSTM model: sequence of last 20 transactions per user as input
- Bidirectional GRU: captures both forward and backward temporal dependencies
- Transformer Encoder: self-attention on transaction sequences (positional encoding)
- Input features per timestep: amount, merchant_category, hour, geo_distance, channel
- Batch normalization + Dropout (0.3) for regularization
- Training: Adam optimizer, lr=1e-3, cosine annealing, early stopping (patience=10)

PyTorch LSTM Bi-GRU Transformer CUDA Weights & Biases

DL-2 Graph Neural Networks — Fraud Rings

Week 4–5

- Build transaction graph: nodes=accounts/merchants, edges=transactions
- Node features: account_age, avg_balance, fraud_history, device_fingerprint
- Edge features: amount, time_delta, channel, shared_IP_flag
- GraphSAGE: inductive learning — works on unseen accounts at inference time
- Graph Attention Network (GAT): attention weights on suspicious connections
- Money laundering detection: layering patterns via multi-hop graph traversal
- Neo4j for graph storage + Cypher queries for pattern investigation

PyTorch Geometric GraphSAGE GAT Neo4j NetworkX DGL

DL-3 Model Explainability (XAI)

Week 5

- SHAP (SHapley Additive Explanations): per-prediction feature importance
- LIME: local surrogate models for individual fraud explanations
- Captum (PyTorch XAI): Integrated Gradients for deep learning models
- GNNExplainer: subgraph explanations for Graph Neural Network decisions
- Counterfactual explanations: what would make this transaction NOT fraud?
- Analyst dashboard: top-3 fraud reasons per alert in plain language

SHAP LIME Captum GNNExplainer Matplotlib Plotly

AI

Track 4: AI / LLM Layer

Week 5–6

AI-1 LLM for Fraud Explanation & Reporting

Week 5

- Claude API integration: auto-explain fraud alerts in Arabic and English
- Prompt engineering: inject SHAP values + transaction context into prompt
- Auto-generate SAR (Suspicious Activity Report) from fraud case data
- RAG pipeline: compliance rules knowledge base (BSP, FATF, AML regulations)
- Document retrieval: LangChain + FAISS vector store on regulatory PDFs
- Output formatting: structured JSON + human-readable summary for analysts

Claude API LangChain FAISS RAG Anthropic SDK OpenAI fallback

AI-2 NLP & Text Analysis

Week 5

- BERT fine-tuned on financial text: intent classification for customer messages
- Phishing detection: classify incoming messages/emails for social engineering
- Named Entity Recognition (NER): extract amounts, account numbers, names
- Sentiment analysis on customer complaints to flag distress signals
- Transaction description parsing: merchant name normalization, MCC mapping
- Multi-language support: Arabic + English NLP pipeline with langdetect

BERT HuggingFace Transformers spaCy fastText langdetect

AI-3 AI Analyst Agent — Autonomous Investigation

Week 5–6

- LangGraph agent: multi-step fraud investigation workflow
- Tools: query_transactions(), lookup_graph(), run_model(), fetch_history()
- Agent autonomously retrieves context, runs analysis, and writes report
- Human-in-the-loop: agent escalates to analyst if confidence < threshold
- Case management integration: auto-create JIRA/ServiceNow tickets
- Audit trail: full agent reasoning log stored for compliance review

LangGraph Claude API Tools API JIRA API FastAPI PostgreSQL

OPS

Track 5: MLOps & Production

Week 5–6

OPS-1 Real-time Scoring Pipeline

Week 5–6

- Apache Flink job: consume Kafka tx stream, enrich with Feature Store, score
- Model serving: NVIDIA Triton Inference Server (XGBoost + PyTorch ensemble)
- Latency target: p99 < 100ms end-to-end (Kafka to decision)
- Decision output: fraud_score (0–1), fraud_label, top_reasons, model_version
- Alerting pipeline: PagerDuty for high-confidence fraud, Slack for summaries
- Fallback rule engine: hard rules for known fraud patterns (velocity > 10 tx/min)

Apache Flink NVIDIA Triton Kafka Streams FastAPI Redis PagerDuty

OPS-2 Model Registry & CI/CD

Week 6

- MLflow Model Registry: stage transitions (Staging → Production → Archived)
- GitHub Actions CI: lint, unit tests, integration tests on every PR
- CD pipeline: auto-deploy to staging, manual approval for production
- A/B testing: shadow mode deployment — new model runs in parallel silently
- Canary deployment: route 5% → 20% → 100% traffic to new model version
- Automated retraining trigger: when PSI (Population Stability Index) > 0.2
- Docker containers + Kubernetes (EKS) for scalable model serving

MLflow GitHub Actions Docker Kubernetes (EKS) ArgoCD Helm

OPS-3 Monitoring, Lineage & Governance

Week 6

- Grafana dashboards: model performance, data quality, pipeline health, latency
- Prometheus + Alertmanager: metrics collection and threshold-based alerts
- Monte Carlo data observability: freshness, volume, schema, distribution monitors
- OpenMetadata: full data lineage from raw source to model prediction
- Data Catalog: all datasets documented with owner, SLA, PII classification
- GDPR compliance: PII masking in logs, right-to-erasure pipeline, audit logs
- Monthly model performance review: drift report, retraining recommendation

Grafana Prometheus Monte Carlo OpenMetadata Great Expectations

08 — Weekly Detailed Plan

Week 1	Foundation & Infrastructure
Day 1–2	Environment setup: Docker, K8s cluster, cloud accounts, repo structure
Day 3–4	Kafka cluster setup, topic design, producer/consumer test with sample data
Day 5–7	S3 bucket structure, Delta Lake setup, Airflow installation & first DAG
✓ Deliverable:	Working ingestion pipeline: raw data flowing from sources into Bronze layer

Week 2	ETL & Feature Engineering
Day 8–9	PySpark ETL: clean transactions, handle nulls, normalize amounts
Day 10–11	dbt models: Silver layer transformations, dim tables, SCD Type 2
Day 12–14	Feature engineering: velocity features, aggregations, merchant risk score
✓ Deliverable:	Gold layer with 50+ features ready; Great Expectations quality report passing

Week 3	Feature Store & Classical ML
Day 15–16	Feast Feature Store: define feature views, materialize offline store
Day 17–18	Baseline models: Logistic Regression, Decision Tree (benchmarks)
Day 19–21	XGBoost + LightGBM: training, Optuna tuning, MLflow experiment tracking
✓ Deliverable:	Best classical model: AUC-ROC > 0.92; MLflow experiments logged

Week 4	Anomaly Detection & Deep Learning
Day 22–23	Isolation Forest + One-Class SVM: unsupervised fraud detection
Day 24–25	LSTM model: sequence encoding of user transaction history
Day 26–28	Graph construction in Neo4j; GraphSAGE training with PyTorch Geometric
✓ Deliverable:	Ensemble model (XGB + LSTM + IsoForest) with higher recall than any single model

Week 5	AI/LLM & Explainability
Day 29–30	SHAP + LIME integration; analyst dashboard with top fraud reasons
Day 31–32	Claude API: fraud explanation prompts, SAR auto-generation
Day 33–35	LangGraph agent: autonomous investigation workflow + tool integration
✓ Deliverable:	AI analyst agent generates full SAR report from fraud alert in <30 seconds

Week 6	MLOps, Production & Handover
Day 36–37	Flink real-time scoring pipeline; Triton model serving; latency testing

Day 38–39	CI/CD pipelines; MLflow registry; Docker + K8s deployment
Day 40–42	Grafana dashboards; OpenMetadata lineage; compliance audit; final review
✓ Deliverable:	Full production system live: real-time scoring at <100ms, all monitors active

09 — Tech Stack Summary

Data Ingestion	Apache Kafka • Apache Airflow • Debezium (CDC) • Avro + Schema Registry • REST / GraphQL APIs
Storage	Amazon S3 / GCS • Delta Lake (ACID) • Snowflake DWH • Apache Iceberg • Redis (online store)
Processing	Apache Spark (PySpark) • dbt Core • Great Expectations • Apache Flink • Pandas / Polars
ML / Classical	XGBoost • LightGBM • Random Forest • scikit-learn • Optuna (tuning) • imbalanced-learn
Anomaly Detection	Isolation Forest • One-Class SVM • DBSCAN • Autoencoder (PyTorch) • Ensemble scoring
Deep Learning	PyTorch • LSTM / Bi-GRU • Transformer Encoder • PyTorch Geometric (GNN) • CUDA / GPU training
Graph	Neo4j (graph DB) • GraphSAGE • Graph Attention Network • NetworkX • DGL (Deep Graph Library)
XAI	SHAP • LIME • Captum • GNNExplainer • Counterfactuals
AI / LLM	Claude API (Anthropic) • LangChain • LangGraph • FAISS (vector store) • HuggingFace Transformers
NLP	BERT (fine-tuned) • spaCy • fastText • langdetect • Named Entity Recognition
Serving	NVIDIA Triton Inference • FastAPI • gRPC • Kubernetes (EKS) • Docker
MLOps	MLflow (tracking + registry) • GitHub Actions (CI/CD) • ArgoCD • Weights & Biases • DVC
Monitoring	Grafana • Prometheus + Alertmanager • Monte Carlo • OpenMetadata • PagerDuty

10 — Deliverables & Success Metrics

Data Pipeline

✓ Kafka + Airflow ingestion pipeline (real-time + batch)
✓ Delta Lake data lake with Bronze / Silver / Gold layers
✓ Feast Feature Store with 50+ engineered features
✓ Great Expectations quality suite — all checks passing

ML Models

✓ Trained ensemble: XGBoost + LSTM + GNN (AUC-ROC > 0.95)
✓ Isolation Forest anomaly detector (F1 > 0.80 on unseen data)
✓ SHAP explainability dashboard for every prediction
✓ MLflow model registry with full experiment history

AI Layer

✓ Claude API integration: fraud explanation in <3 seconds
✓ Auto-generated SAR reports (PDF output, bilingual)
✓ LangGraph agent: autonomous case investigation
✓ RAG knowledge base: AML/compliance regulations indexed

Production System

✓ Real-time scoring pipeline: p99 latency < 100ms
✓ CI/CD pipeline: automated testing + deployment
✓ Grafana monitoring: 20+ dashboards (model, data, infra)
✓ OpenMetadata: full lineage for every dataset and model

Success Metrics

Metric	Target	Track
AUC-ROC (ensemble)	**> 0.95**	**ML**
Precision @ 90% Recall	**> 85%**	**ML**
Real-time latency (p99)	**< 100ms**	**MLOps**
Daily pipeline SLA	**< 30 min**	**DE**
SAR generation time	**< 30 sec**	**AI**
Data quality score	**> 99%**	**DE**
Model drift trigger (PSI)	**< 0.2**	**MLOps**
False positive rate	**< 5%**	**ML**

This document represents the complete 6-week project plan for the Financial Fraud Detection System. All tracks run in parallel with clear weekly checkpoints and measurable deliverables. The system is designed for production scalability, regulatory compliance, and continuous improvement.